XIAOYING SONG

xiaoyingsong@my.unt.edu Homepage LinkedIn Google Scholar

RESEARCH INTERESTS

Human-Agents Collaboration, Personalized Alignment, Natural Language Processing

EDUCATION

University of North Texas, USA *August 2023 - Now*
PhD Candidate in Information Science (Concentration in Data Science)

Central China Normal University, China *August 2021 - June 2023*
Master in Information Science

PUBLICATIONS

1. Tan, Q., **Song, X.**, Akbari, A., Akbari, A., Wang, Y., Zhai, X., . . . & Yuan, G. Palette: A modular, controllable, and efficient framework for on-demand authorized safety alignment relaxation in LLMs. *arXiv preprint arXiv:2605.24154*, 2026.
2. Tan, Q., **Song, X.**, Cheng, N., Liu, N., Zhai, X., Hong, L., . . . & Yuan, G. Q-Realign: Piggybacking realignment on quantization for safe and efficient LLM deployment. *arXiv preprint arXiv:2601.08089*, 2026.
3. **Song, X.**, Luo, P., Thomale, J., Zavalina, O., & Hong, L. Comparative analysis of large language models' performance in book classification tasks using Library of Congress Classification system. *Journal of Information Science*, 2026.
4. **Song, X.**, Anik, A. S., Barua, D., Luo, P., Ding, J., & Hong, L. Speaking at the right level: Literacy-controlled counterspeech generation with RAG-RL. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing **(EMNLP)***.
5. **Song, X.**, Anik, A. S., Blanco, E., Frías-Martínez, V., & Hong, L. A dynamic fusion model for consistent crisis response. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing **(EMNLP)***.
6. Anik, A. S.*, **Song, X.***, Wang, E., Wang, B., Yarimbas, B., & Hong, L. Multi-agent retrieval-augmented framework for evidence-based counterspeech against health misinformation. In *Proceedings of the Second Conference on Language Modeling (COLM)*, 2025. (* Co-first authors)
7. **Song, X.**, Perez, S. L., Yu, X., Blanco, E., & Hong, L. Echoes of discord: Forecasting hater reactions to counterspeech. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics **(NAACL)***.
8. **Song, X.**, Mamidisetty, S., Blanco, E., & Hong, L. Assessing the human likeness of AI-generated counterspeech. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*, 2025.
9. **Hong, L.**, Luo, P., Blanco, E., & Song, X. Outcome-constrained large language models for countering hate speech. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing **(EMNLP)***.
10. **Song, X.**, Anik, A. S., Frías-Martínez, V., & Hong, L. Dynamic fusion of large language models for crisis communication. In *Proceedings of ISCRAM*, 2025.
11. Liu, J., **Song, X.**, Zhang, D., Thomale, J., He, D., & Hong, L. A hybrid framework for subject analysis: Integrating embedding-based regression models with large language models. In *Proceedings of ASIS&T*, 2025.
12. Perez, S. L., **Song, X.**, & Hong, L. Analyzing the language of rejection: A study of user flagging responses to hate speech on Reddit. *Information Research*, 2025.
13. Tan, Q., **Song, X.**, Ye, G., & Wu, C. An effective negative sampling approach for contrastive learning of sentence embedding. *Machine Learning*, 2023.

RESEARCH PROJECTS

User-Centered Multi-Turn Conversation Optimization

Aug 2025 – Present
Project Leader

- Simulate multi-turn conversations between users and agents to model interactive decision-making and user-centered information-seeking behavior.
- Design reward functions that balance clarification-seeking behavior, response quality, and communication cost.
- Optimize dialogue systems using GRPO to improve adaptive questioning and multi-turn intervention strategies.
- **Research Outcomes:** One study under review, one study is in progress.

Knowledge-Enhanced Crisis Response and Misinformation Mitigation

Aug 2024 – Present
Project Leader

- Build retrieval-augmented generation systems grounded in authoritative humanitarian resources to improve responses to natural crises and crisis-related misinformation.
- Construct hierarchical knowledge graphs to organize external knowledge for users with diverse backgrounds and information needs.
- Develop personalized response-generation methods that integrate user context and verified external knowledge.
- **Research Outcomes:** Two studies published in EMNLP 2025, and two studies are in progress.

Spatiotemporal Reasoning in Large Language Models

Aug 2025 – Present
Project Leader

- Develop benchmarks for evaluating LLM reasoning on spatiotemporally conditioned crisis needs and situational awareness.
- Analyze how LLMs interpret location, time, evolving crisis contexts, and resource constraints in crisis-oriented decision-making tasks.
- Enhance spatiotemporal reasoning through multi-agent collaboration for crisis-oriented decision support.
- **Research Outcomes:** Two studies are under review, one study is in progress.

HONORS AND AWARDS

Second Place, CLEF 2026 CheckThat! Lab Competition	*2026*
Honorable Mention, ASIS&T SIG-SM Student Competition	*2026*
First Place, UNT Phil and Lis Turner Outstanding Award	*2024–2025*
UNT Porter-Evans Scholarship	*2024–2025*
UNT COI/IS Leon M. Liddell Scholarship	*2023–2024*
UNT Graduate School Travel Award	*2023–2024*
CCNU Excellent Student Awards	*2021–2023*
CCNU Graduate Scholarships	*2021–2023*

EXPERIENCES

Research Assistant — *CCNU, UNT, Aug 2021 – Present*

Teaching Assistant — *UNT, Aug 2023 – Jul 2024*

Introduction to Data Science; Data Visualization; Information Architecture

Conference Tutorials

- **Song, X**. & Hong, L. (2025, 2026). *AI-assisted Communications*.

- **Song, X**. & Hong, L. (2024). *Data Collection, Classification, and Transformation*.
- **Song, X**. & Hong, L. (2024). *Comprehending the Semantic Essence of Text: Leveraging SEANCE for Textual Analysis*.

All presented at IDEA Institute on AI, Association for Information Science & Technology (ASIS&T).

MISCELLANEOUS

Programming & Data Processing: Python (Pandas, NumPy, PySpark); proficient in data wrangling, preprocessing, and scalable pipeline development.

Machine Learning & AI: Hands-on work in setting up, fine-tuning, and evaluating LLMs using Hugging Face Transformers and PEFT methods.

Database Management: Proficient in SQL; experienced with MySQL and Toad for relational data querying and manipulation.

Academic Service: Reviewer for conferences ACL, NAACL, EMNLP, COLING, COLM, and NeurIPS